

Tien Phu Tran

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Profile Summary

Computer Science graduate specializing in Artificial Intelligence and Machine Learning. Proven experience in deep learning, architecting NLP pipelines, GANs, Computer Vision, and backend architectures using Python, PyTorch, and Django. Seeking ML Engineer, Data Scientist, or Software Developer roles.

Education

University of Houston, B.S. in Computer Science – GPA: 3.9 June 2023 – December 2025

- **Relevant Coursework:** Artificial Intelligence, Natural Language Processing, Medical Image, Digital Image Processing, Data Science, Algorithms, Database, Web Development.

Houston Community College, A.S. in Computer Science – GPA: 4.0 August 2020 – May 2023

Projects

Sentiment Analysis for Amazon Search Engine | Personal Project Sep 2025 - Dec 2025

- Designed an information retrieval system over 200,000+ Amazon reviews to extract aspect-opinion pairs, migrating from a baseline Boolean to a deep semantic search.
- Built an NLP pipeline featuring text preprocessing, inverted index, lexicon-based sentiment polarity, and cosine similarity-based topic expansion for aspect-opinion query.
- Improved retrieval quality by using SBERT and BGE ranking models with a custom Rank Penalty algorithm, achieving 86% of peak system accuracy.
- Tools Used: Python, PyTorch, Pandas, NLTK, Regex, SpaCy, SBERT, Hugging Face, pickle, Google Colab.

MRI-to-CT Image Translation using GANs (Link) | Team Project Sep 2025 - Dec 2025

- Implemented Pix2Pix model in PyTorch using an 8-level U-Net generator for MRI-to-CT conversion and a 70×70 PatchGAN discriminator for adversarial learning between real and converted.
- Developed a preprocessing pipeline for 1,250 paired slices from the CHAOS/IXI datasets, including spatial alignment, flips, rotations, and intensity normalization to increase data diversity and robustness.
- Trained and optimized the model over 100 epochs, achieving 18.83 dB PSNR while evaluating model performance using SSIM, PSNR, and MAE metrics.
- Tools Used: PyTorch, Python, NumPy, Pandas, Scikit-Image, OpenCV, Google Colab.

Document Approval & User Management System (Link) | Team Project Jan 2025 - May 2025

- Developed a full-stack Django application for document approval workflows, implementing system logging, role-based access, and a dynamic assignment hierarchy for multi-level task delegation.
- Designed a backend pipeline utilizing a LaTeX rendering system and PostgreSQL to systematically store structured data and generate static PDFs.
- Built a React frontend to deliver a dynamic user interface for managing workflow actions and submissions.
- Tools Used: Django, React, PostgreSQL, LaTeX, Python, JavaScript, HTML/CSS, Docker.

Technical Skills

Languages: Python, PowerShell, Bash, SQL, C++, JavaScript, HTML/CSS

Machine Learning / AI: TensorFlow, PyTorch, OpenCV, Scikit-Image, Hugging Face Transformers, LangChain, CNNs, GANs (Pix2Pix, CycleGAN), Natural Language Processing, Deep Learning

Data & MLOps: Pandas, NumPy, Matplotlib, MLflow, Jupyter Notebook, ETL Pipelines, Regex

Software & Web Dev: Django, React, MySQL, PostgreSQL, REST APIs, LaTeX